Bereich: XML (1)

Google-Maps-Gazetteer-Service	**Musterlösung**
Package: `de.dhbwka.java.exercise.xml`	**Klasse:** `Gazetteer`

```java
package de.dhbwka.java.exercise.xml;

/**
 * Part of lectures on 'Programming in Java'. Baden-Wuerttemberg
 * Cooperative State University.
 *
 * (C) 2016 by W. Geiger, T. Schlachter, C. Schmitt, W. Suess
 *
 * @author DHBW lecturer
 * @version 1.0
 */
public class Bounds {
    public double south, north, west, east;

    public Bounds() {}

    public Bounds(String east, String north, String west, String south) {
        this(Double.parseDouble(east), Double.parseDouble(north),
             Double.parseDouble(west), Double.parseDouble(south));
    }

    public Bounds(double east, double north, double west, double south) {
        this.east = east;
        this.north = north;
        this.west = west;
        this.south = south;
    }

    @Override
    public String toString() {
        return "Bounds: (east="+east+", north="+north+", west="+west+
                        ", south="+south+")";
    }

}
```

```java
package de.dhbwka.java.exercise.xml;

import java.io.IOException;
import javax.xml.parsers.DocumentBuilder;
import javax.xml.parsers.DocumentBuilderFactory;
import javax.xml.parsers.ParserConfigurationException;
import org.w3c.dom.DOMException;
import org.w3c.dom.Document;
import org.w3c.dom.Node;
import org.w3c.dom.NodeList;
import org.xml.sax.SAXException;

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 */
public class Gazetteer {

    public static Bounds getBounds(String queryTerm) {
        Bounds result = new Bounds();
        try {
            DocumentBuilderFactory factory =
DocumentBuilderFactory.newInstance();
            DocumentBuilder parser = factory.newDocumentBuilder();
            String url =
"https://maps.googleapis.com/maps/api/geocode/xml?sensor=false&address="+queryTerm;
            Document doc = parser.parse(url);
            // Auf (einzige) formatted_address setzen und...
            String form_adress = doc.getDocumentElement()
                .getElementsByTagName("formatted_address")
                .item(0).getTextContent();
            System.out.println("Adresse: "+form_adress);

            String long_name = "kein locality-Name gefunden";
            NodeList nl = doc.getDocumentElement()
                            .getElementsByTagName("address_component");
                            // evtl. mehrfach vorhanden!
            for (int i=0; i<nl.getLength(); i++) {
                if (getNamedChildNode(nl.item(i),
                    "type").getTextContent().equals("locality")) {
                    long_name = getNamedChildNode(nl.item(i),
                                "long_name").getTextContent();
                    break;
                }
            }
            System.out.println("Long Name: "+long_name);

            //Lage
            Node loc = doc.getDocumentElement().getElementsByTagName("location")
                    .item(0);
            String lat = getNamedChildNode(loc, "lat").getTextContent();
            String lng = getNamedChildNode(loc, "lng").getTextContent();
```

```
            System.out.println("Location: (latitude=" + lat + ""
                    + ", longitude=" + lng + ")");

            Node bounds = doc.getDocumentElement().getElementsByTagName("bounds")
                            .item(0);
            result = new Bounds(
                    getNamedChildNode(getNamedChildNode(bounds,
                      "northeast"),"lng").getTextContent(), // east
                    getNamedChildNode(getNamedChildNode(bounds,
                      "northeast"),"lat").getTextContent(), // north
                    getNamedChildNode(getNamedChildNode(bounds,
                      "southwest"),"lng").getTextContent(), // west
                    getNamedChildNode(getNamedChildNode(bounds,
                      "southwest"),"lat").getTextContent()  // south
                    );
        } catch (ParserConfigurationException | SAXException | IOException
                | DOMException ex) {
            System.err.println("Fehler bei getBounds: "+ex.getMessage());
        }
        return result;
    }

    // Liefert den ersten Kind-Koten mit dem angegebenen Namen
    public static Node getNamedChildNode(Node n, String name) {
        NodeList children = n.getChildNodes();
        for(int i=0; i<children.getLength(); i++)
            if (children.item(i).getNodeName().equals(name))
                return children.item(i);
        return null;
    }

    public static void main(String[] args) {
        System.out.println(getBounds("Karlsruhe"));
    }
}
```